Day 4 Task — CyberChef Lab: Answer Sheet

Tekloomy Cyber Security Internship · Module 1: Zero to Defender · Day 4

Part A — Decode the hidden messages

#	Decoded message	Operation(s) used
A1	Welcome to Tekloomy. Encoding is not encryption.	From Base64
A2	Hash everything you download.	From Hex
A3	Zero to Defender	ROT13 (Caesar shift of 13)
A4	Backups beat ransomware	From Base64 → From Hex (Base64 gives 4261636b7570…, Hex gives the sentence)

Part B — Hashing (SHA-256)

Input (type exactly)	First 8 chars of SHA-256
tekloomy	`654b8d2e`
Tekloomy	`53c9cd59`
tekloomy (with a space at the end)	`f6636caf`

B4. Changing one capital letter changed the entire hash (it looks completely unrelated; this is the avalanche effect). So even a 1-bit tampering with a downloaded file gives a different hash, and comparing it with the publisher's hash proves the file is intact and unmodified.

Part C — Inspect a real certificate

This part depends on what your browser shows today, so confirm each value on your own screen. Typical values for www.google.com are given as a guide.

Field	What you found
Issued to (Common Name / domain)	www.google.com
Issued by (Certificate Authority)	Google Trust Services (an intermediate such as WR2; check yours)
Valid until (expiry date)	Read from your certificate viewer (Google certs last about 3 months)
Public key algorithm and size	Elliptic Curve (ECC), P-256 (256-bit), typically; check yours

C5. expired.badssl.com shows “Your connection is not private” (NET::ERR_CERT_DATE_INVALID). It means the site's certificate has passed its expiry date, so the browser can no longer trust that the server is who it claims to be, and traffic could be intercepted. No, do not click Proceed: especially never enter passwords or personal data on such a site.

Part D — Encrypt for real (AES, CBC)

Key (hex) `00112233445566778899aabbccddeeff`, IV (hex) `000102030405060708090a0b0c0d0e0f`.
Recipe: AES Encrypt (Key/IV type = Hex, Mode CBC, Input Raw, Output Hex), then AES Decrypt with the same settings.

D1. The output depends on your first name. Worked example for the name **Aarav**:
`Encrypted (hex): 41578b6103fa6c1de51abf2b071f1362`
Decrypt with same key/IV returns: Aarav
Write your own name's output from CyberChef here: _____

D2. Changing even one character of the key makes decryption fail: CyberChef shows an error or garbled, unreadable bytes instead of your name. For ransomware this matters because the data is only recoverable with the exact key: without it, the encrypted files are useless, which is why attackers hold the key for ransom and why offline backups are the real defence.

Tip: do the CyberChef steps yourself once so you can explain them in the Day 5 session.